

RADWIN

TerraNet Installation Guide

Release 1.13.00

Document Version 1.3



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1 Introduction

1.1 Document Scope

This document describes how to install the TERRANET Portfolio Outdoor Units (ODUs).

1.2 Revision History

	Date	Document Revision	SW Release	Revision details
1.	May 2024	1.0	1.0.00	Initial Revision
2.	February 2025	1.1	1.13.00	- Add V90 support - Add new mounting kits
3.	April 2025	1.2	1.14.00	Add sealing kit usage description
4.	June 2025	1.3	1.14.00	- Add Advanced Mounting Kit details - Add vertical and horizontal antenna positioning

1.3 What's New in this Release

2 new mounting kits have been added for TerraNet products:

- Standard: Basic RADWIN mounting kit
- Advanced: one degree step mounting kit for PtP purposes
- Add TerraNet V90 and its antenna lenses
- Add Sealing Kit usage for TerraNet V40 and V120

1.4 Product Family Overview

The TerraNet Portfolio delivers up to 2.3Gbps (depending on the regulation) in a point-to-multi-point or point-to-point architecture and is the ideal choice for smart cities last mile connectivity and for LOS Gigabit backhaul.

The TerraNet Portfolio includes the following models:

RADWIN 2000-E Family Model Comparisons

Model Name	PN	Product Name	Max Throughput	Form Factor
V120	RW-6V02-9160	RW6000/V120/F60/UNI/INT	2.3Gbps	Integrated
	RW-6V02-I160	RW6000/V120/F60/LIM/INT		
	RW-6V02-J160	RW6000/V120/F60/JPN/INT		
V40	RW-6N02-9160	RW6000/V40/F60/UNI/INT	2.3Gbps	Integrated
	RW-6N02-I160	RW6000/V40/F60/LIM/INT		
	RW-6N02-J160	RW6000/V40/F60/JPN/INT		

Model Name	PN	Product Name	Max Throughput	Form Factor
V90	RW-6P02-9260	RW6000/V90/F60/UNI/INT	2.0Gbps	Integrated and External Antenna Lenses
	RW-6P02-I260	RW6000/V90/F60/LIM/INT		
	RW-6P02-J260	RW6000/V90/F60/JPN/INT		



Some options and models may not be available for your regulatory environment.

1.5 Safety and Security

1.5.1 Safety

Before working on equipment connected to power lines or telecommunication lines, remove jewelry or any other metallic object that may encounter energized parts.

Use extreme care when working at heights.

All RADWIN products should be grounded during operation.

The use of lightning protection is dependent on regulatory and end-user requirements.

To protect against overexposure to RF energy, all persons should maintain safe distances from radio sources.

When the system is operational, avoid standing directly in front of the antenna. Strong RF fields are present when the transmitter is on.

1.5.2 Security Recommendations

Change the default user password and set a new link password.

Use only SNMPv3 for monitoring and disable SNMPv2.

1.6 Modes of Operation

TerraNet products could be configured for Point-to-Multi-Point and Point-to-Point operations. In Point-to-Multi-Point, the Outdoor Base Station could connect to up to 32 Clients.

1.6.1 Point-to-Multi-Point

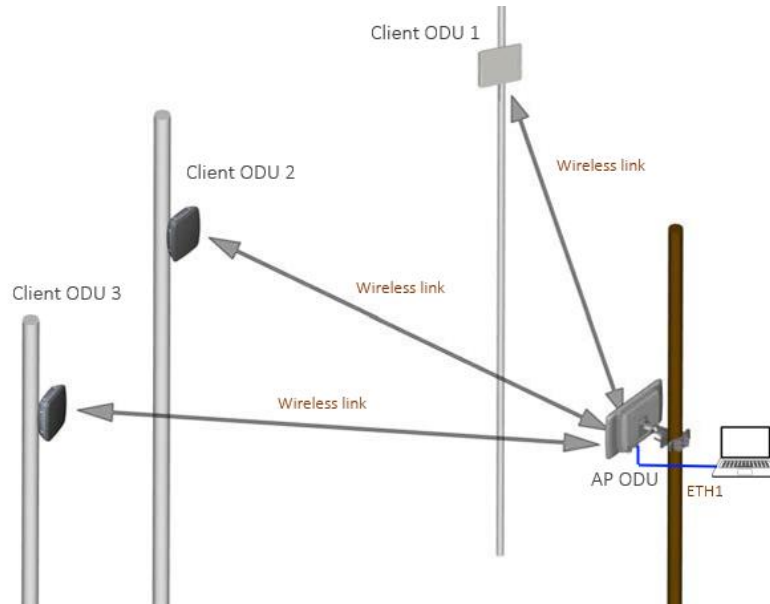


Figure 1: PtMP Connection Scheme

Any TerraNet ODU can be configured either as Access Point (AP) or as a Station (Client). In Point-to-Multi-Point Access Point mode, up to 32 Stations can be connected.

1.6.2 Point-to-Point

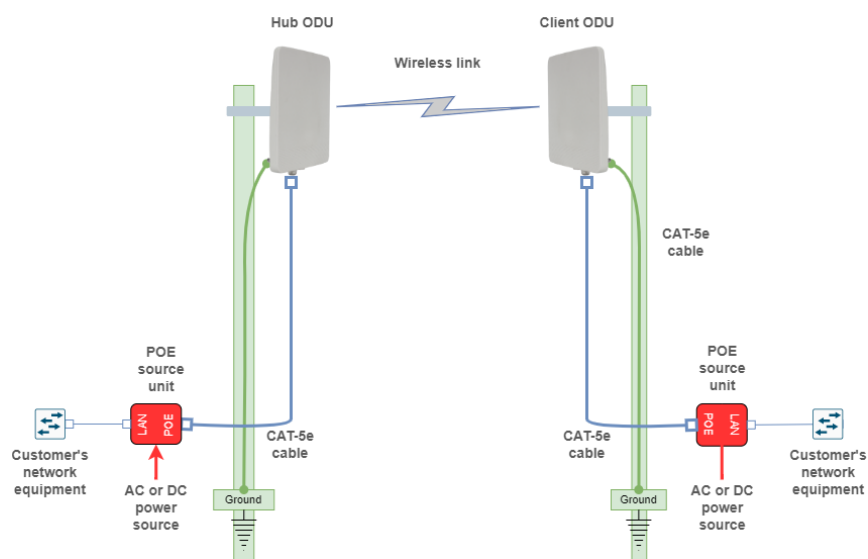


Figure 2: PtP Connection Scheme



For simplicity, Lightning Protection Units (LPUs) are not shown in the following figure **but are recommended**.

Any TerraNet ODU in a PTP link can be configured either as PTP Access Point or as PTP Station.

Both for PTMP and for PTP, an Access Point will transmit a beacon on the selected frequency channel, and a Station will scan for the beacon and connect to the Access Point.



Radwin recommends that any ODU connected to the operator's backbone network will be configured as an Access Point, while ODU's at remote sites (closer to the end customer) will be configured as a Station.

The differences between Hub and Client are summarized in the following table:

Overview of the Differences between AP and Client

AP	Client
Transmits beacon, waiting for Client connection	Searches for beacon and establishes a connection to the Hub
Contains some master settings for all devices	Receives settings from the Hub on link establishment

2 Box Content



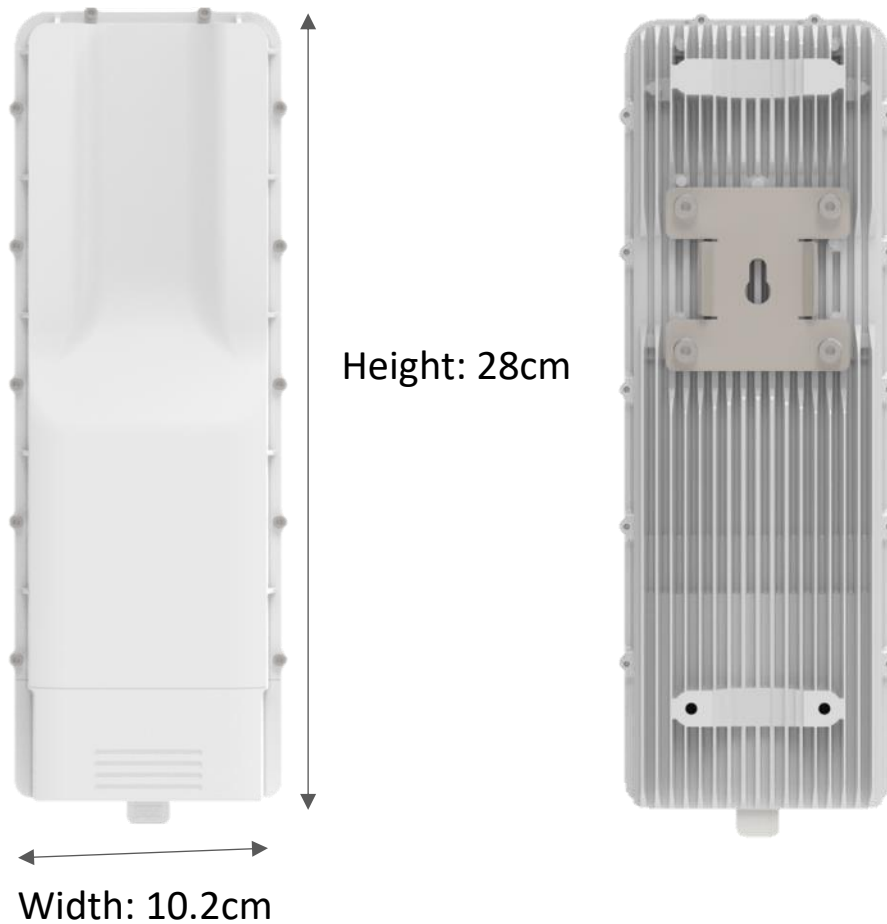
Please consider recycling our eco-friendly product packaging.

Item	Qty
TerraNet Radio Unit w/ attached mounting back plate	1
48V .5A PoE adapter and power cable	1

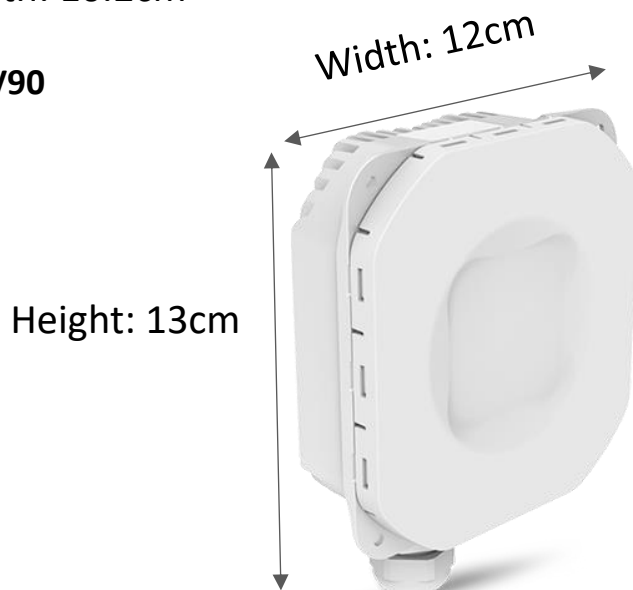
3 Installation

3.1 Mechanical Specifications

3.1.1 V40 and V120



3.1.2 V90



3.2 Interfaces

3.2.1 V40 and V120

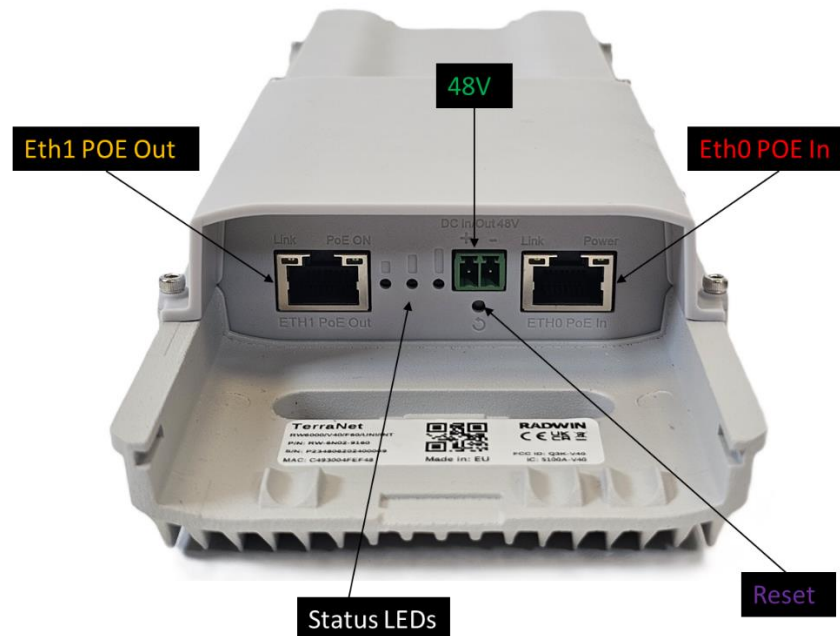


Figure 3: V40 and V120 Interfaces

TerraNet device comes with multiple interfaces:

- Eth0 POE In (2.5Gbps)
 - Included POE unit
 - RADWIN POE and IDU-S/SI
 - 802.3at Class 4 (30W)
- Eth1 POE Out (1Gbps)
 - RADWIN SU PRO / SU AIR / Alpha
 - Other similar passive POE devices (example: IP cameras)
 - Pinout: (+) pins 4/5 (-) pins 7/8
 - Up to <13W
- Direct 48V DC input
 - Screw terminal connector is not included
 - Supports only +48V (negative ground)
 - Enables POE out

- Reset button
 - Short press = reboot
 - Long press >20 seconds = factory default
- Status LEDs
 - For more explanations about LEDs, please see chapter 3.5.

3.2.2 V90

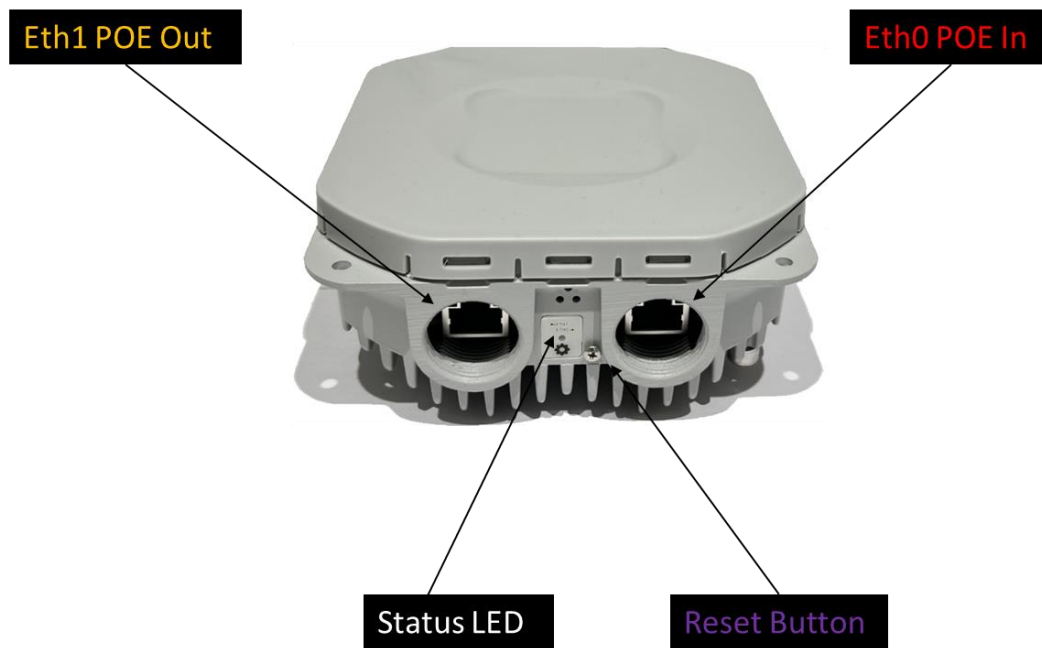


Figure 4: V90 Interfaces

TerraNet device comes with multiple interfaces:

- Eth0 POE In (2.5Gbps)
 - Included POE unit
 - RADWIN POE and IDU-S/SI
 - 802.3at Class 4 (30W)
- Eth1 POE Out (1Gbps)
 - RADWIN SU PRO / SU AIR / Alpha
 - Other similar passive POE devices (example: IP cameras)
 - Pinout: (+) pins 4/5 (-) pins 7/8
 - Up to <13W
- Reset button

- The button is located under the screw
 - Short press = reboot
 - Long press >20 seconds = factory default
- Status LED
 - For more explanations about LED, please see chapter 3.5.

3.3 Mounting the device

You could mount the device on a wall or a pole using the provided mounting kit.

Both mounting kits are compatible with all our TerraNet devices: V40, V90 and V120.

Each mounting kit has its advantages and disadvantages. In the table below, please find our recommendation for different use cases.

<i>Use Case</i>	<i>Recommended Mounting Kit</i>
<i>Deployment on high tower for PtMP BS</i>	Standard MK
<i>V90 Client deployment with huge difference in elevation between the BS and the SU</i>	Standard MK
<i>V90 Client deployment with small difference in elevation between the BS and the SU</i>	Advanced MK
<i>Backhaul deployment</i>	Advanced MK on both side

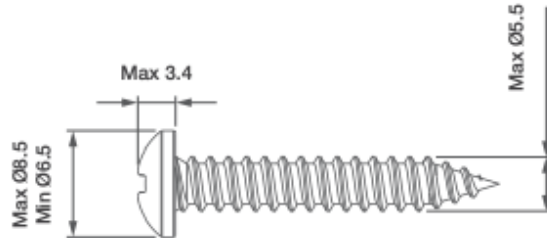
3.3.1 Prepare the tools

Depending on the mounting kit you planned to use and where you plan to mount the device, few tools could be required:

- Crimping tool
- Spanner/wrench
- Cable ties
- Drill and appropriate bits to make holes for PoE and bolts

3.3.2 Screw Specification

Screw specification (mm):



3.3.3 Wall Mounting Instructions

Wall Mount without Mounting Kit – Only for V40 and V120

When using built-in mounting plate:

- Alignment is not possible
- 3x holes are required

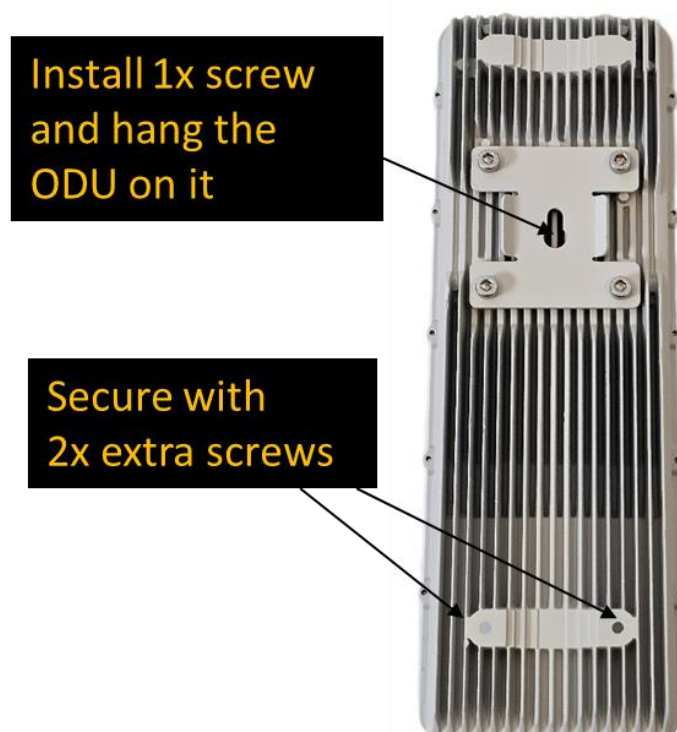


Figure 5: Attach the device to a wall

Here is a description of the process when fixing on a wall.

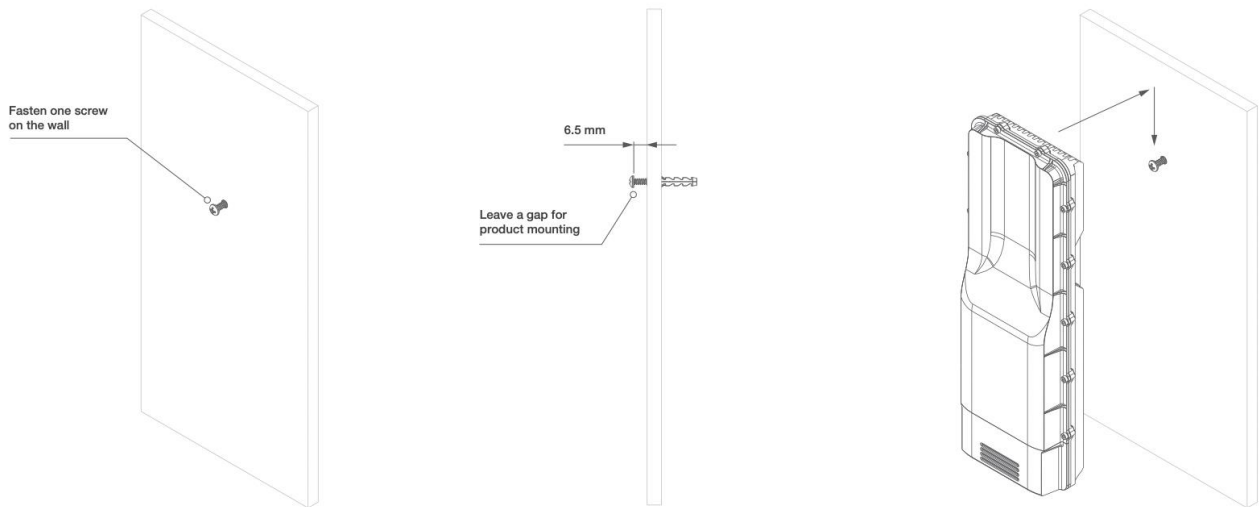


Figure 6: Use a screw to mount the device on a wall

Wall Mount using the Basic Mounting Kit

For a more secure mounting on a wall, the basic mounting kit could be used.

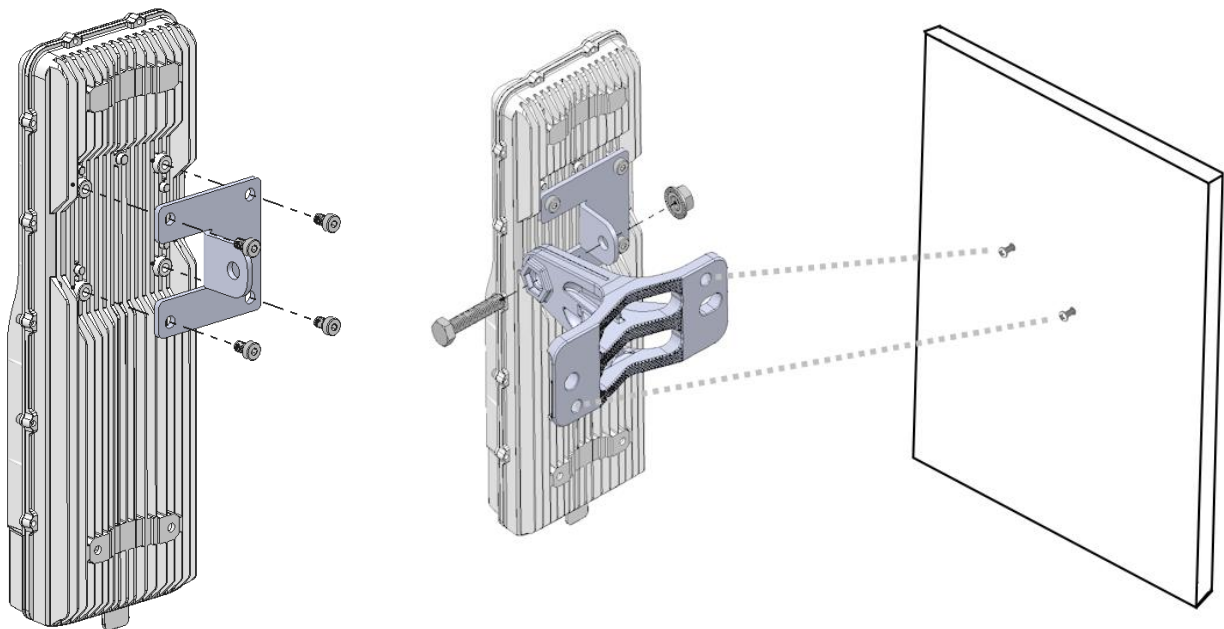


Figure 7: Use the basic mounting kit to mount the device on a wall

1. Use the TerraNet plate and fasten it to the TerraNet radio with the supplied screws

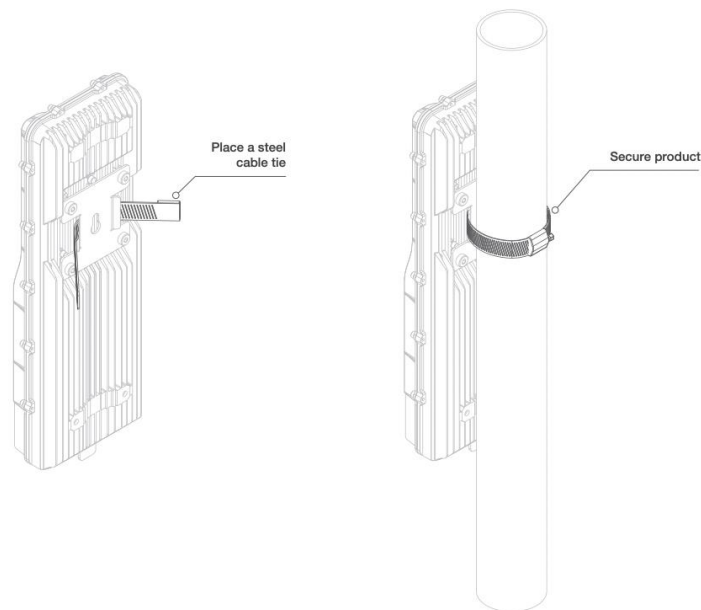
2. Fasten the radio holder to the protruding lip of the TerraNet plate
3. Attach the assembly to the wall as described above using 2 screws

3.3.4 Pole Mounting Instructions

Using the metal band tie

Here are the steps to follow to mount the device on a pole:

- Release the metal band



- Attach the ODU to the pole

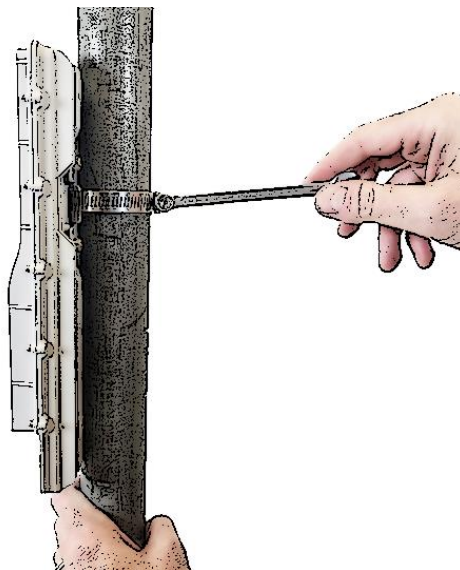


Figure 8: Attach the device on the pole

- Rotate around the pole to roughly align the center azimuth by eye / compass

Using the Basic Mounting Kit

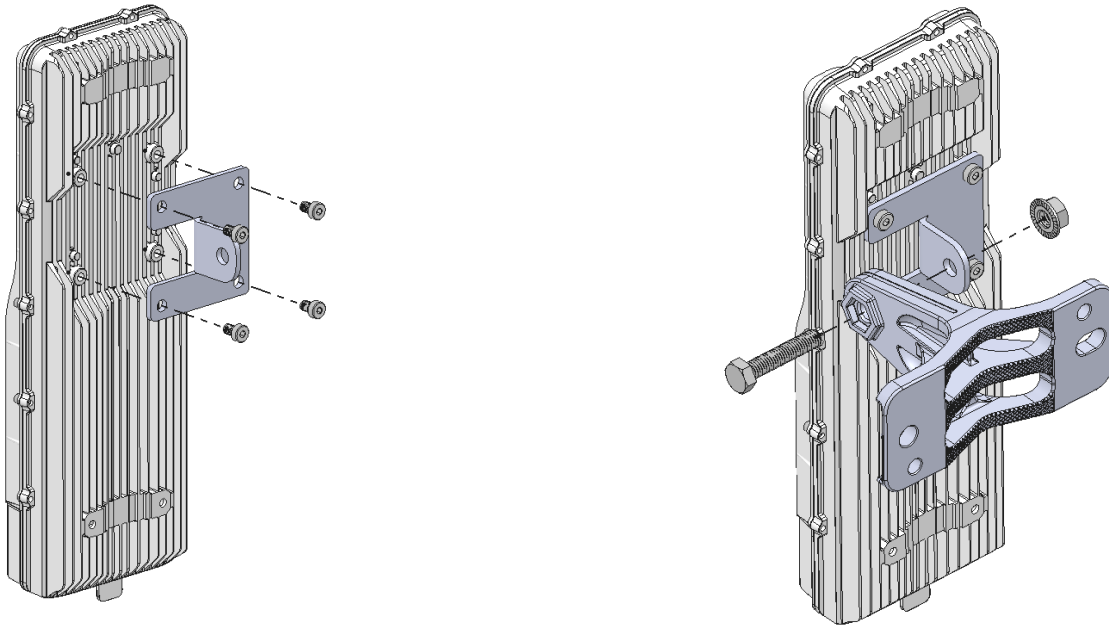


Figure 9: Prepare the TerraNet device

1. Use the TerraNet plate and fasten it to the TerraNet radio with the supplied screws by using a torque of 8Nm
2. Fasten the radio holder to the protruding lip of the TerraNet plate

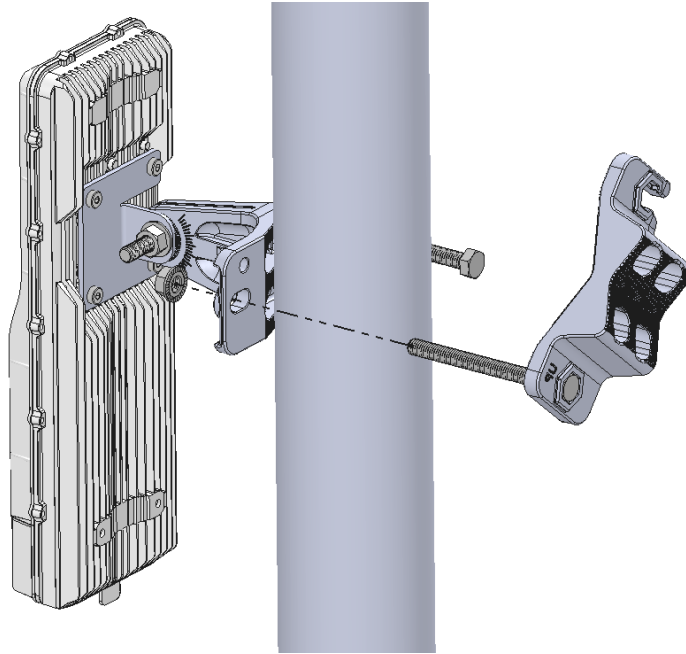
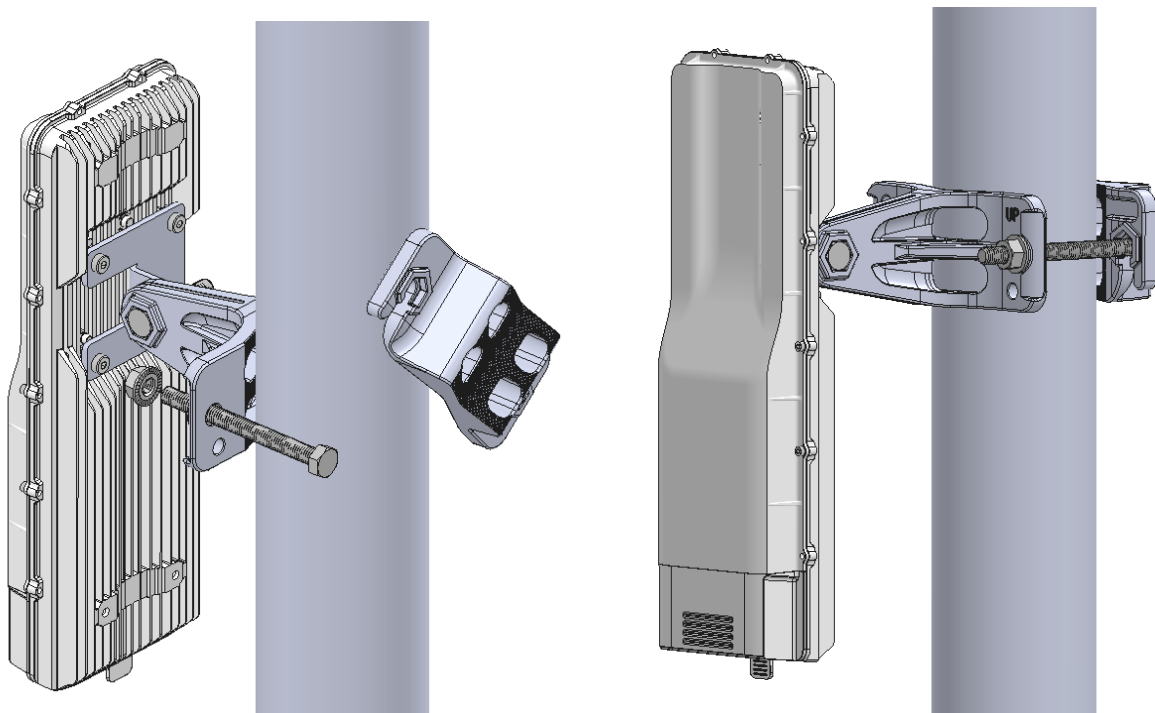


Figure 10: Mount the TerraNet device on the pole

3. Place the assembly around the pole, then fasten the pole clamp to the radio holder



4. Move the device to the correct azimuth, then secure the device on the pole by applying a torque of 12Nm on both screws

Using the Advanced Mounting Kit

40-50 mm Poles

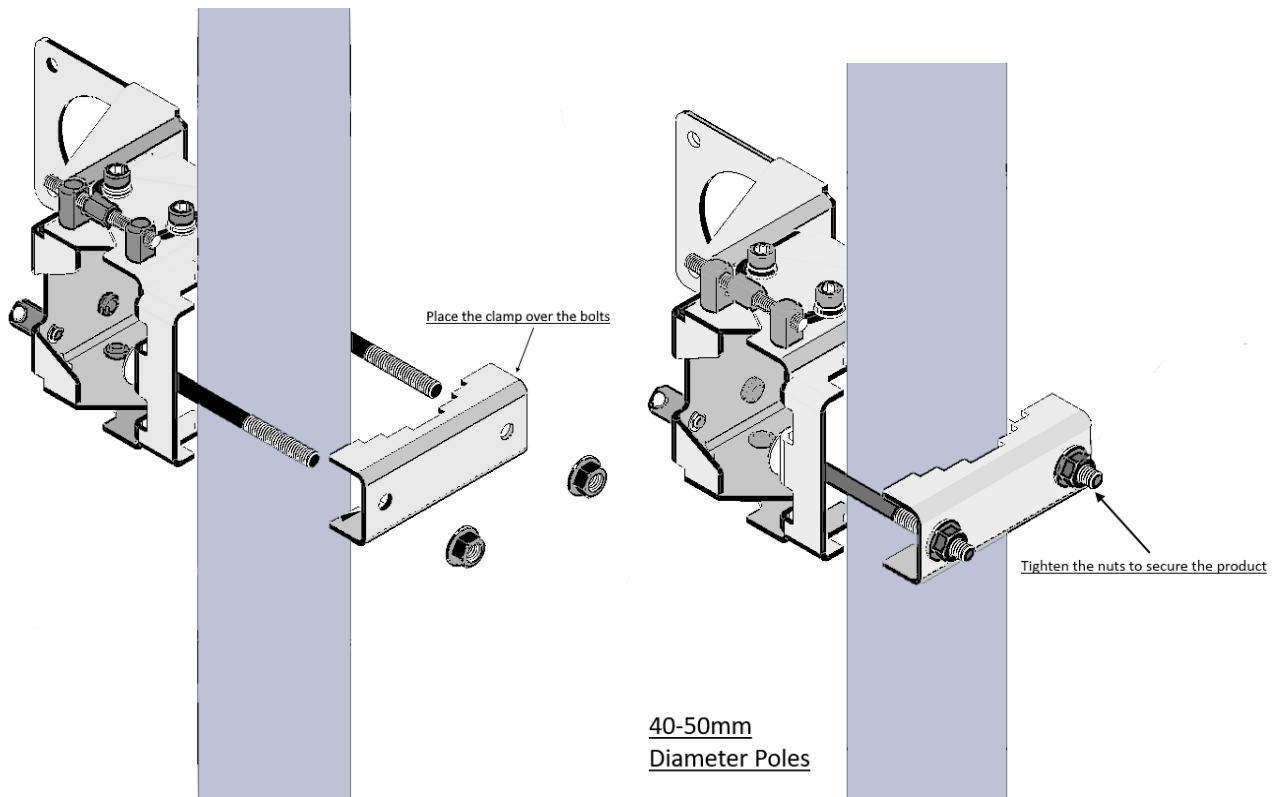


Figure 11: Mount the TerraNet device on a 40-50mm pole

1. Mount the advanced mounting kit on the pole
2. Place the clamp over the bolts
3. Tighten the nuts to secure the product by using a torque of 10Nm

50-60 mm Poles

1. Remove the bolts from the advanced mounting kit
2. Place the steel cable band ties
3. Mount the advanced mounting kit on the pole
4. Secure the product by tightening the steel cable band ties on the pole using a torque of 7Nm

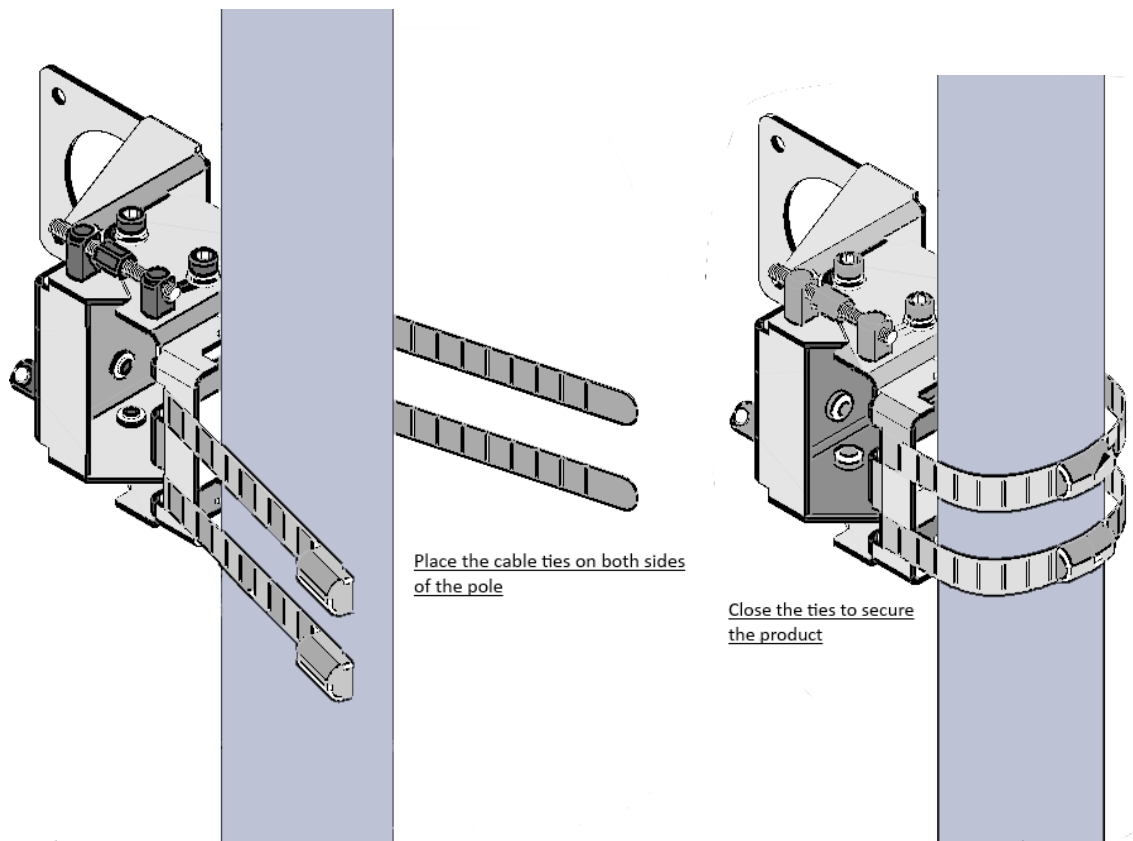


Figure 12: Mount the TerraNet device on a 50-60mm pole

When the advanced mounting is secured on the pole, you could proceed with the radio unit.

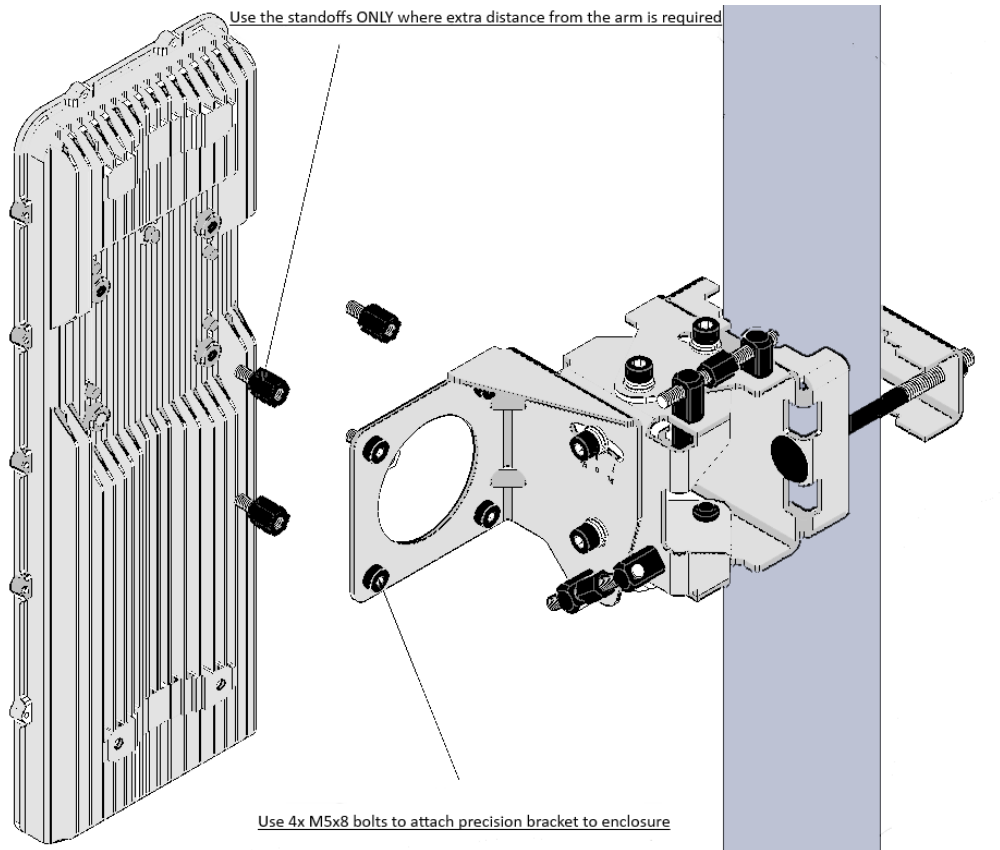


Figure 13: Set the TerraNet device on the advanced mounting kit

If required you could use the standoffs provided in the advanced mounting kit, then secure them using a torque of 8Nm.

The attachment to the precision bracket is done using 4x M5x8 bolts, the torque used should be 8Nm.

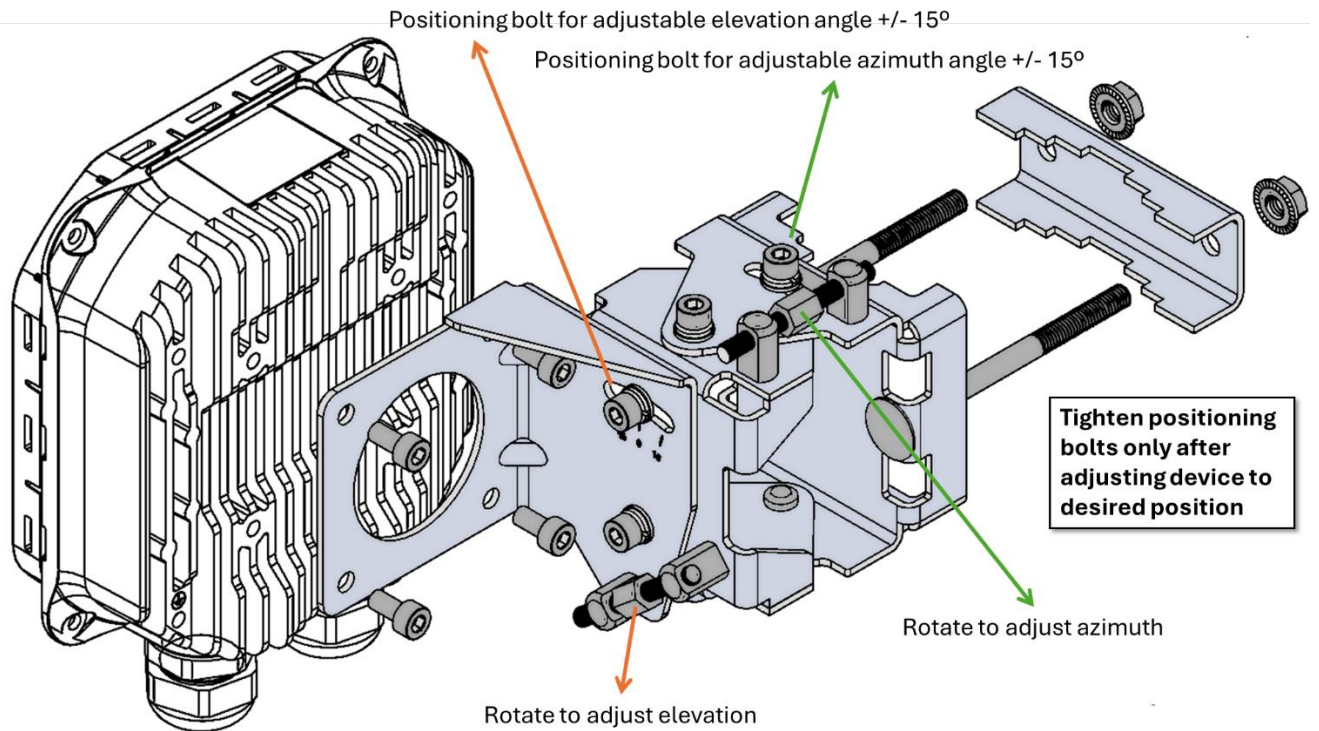
Then make azimuth and elevation correction, when correctly set secure the 6x M5 bolts of the advanced mounting kit using a torque of 8Nm.

Adjusting the advanced mounting kit

The advanced mounting kit could be fine adjusted for both azimuth and elevation. See the picture below to learn about these adjustments.



Do not use power tools to tighten precision bracket, you may permanently damage the bracket itself, and the integrity of the precision bracket will be compromised.



3.3.5 Connect an external antenna to Terranet V90

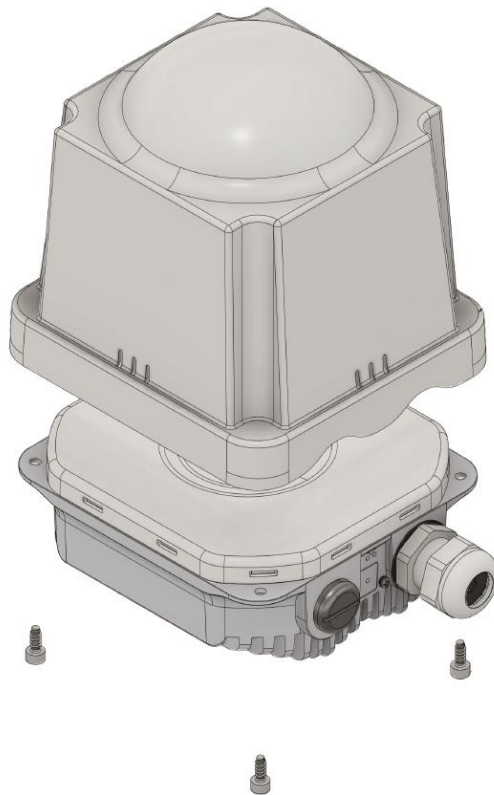


Figure 14: Add an external antenna to TerraNet V90 device

1. Check that the antenna is correctly positioned, the water extraction holes should be on the bottom and on the side of the device. **No water evacuation holes should be visible on top.**

Please be careful to mount correctly the antenna on the device to avoid huge performance degradation in case of rain.

2. Secure the external antenna on the TerraNet V90 body using the 4 provided M4x10 screws. The torque applied should be of 1.5Nm maximum



Using a torque > 1.5Nm could damage the holes in the plastic of the external antenna and cause fragility issues.

Vertical and Horizontal use with 90° and 45° antennas

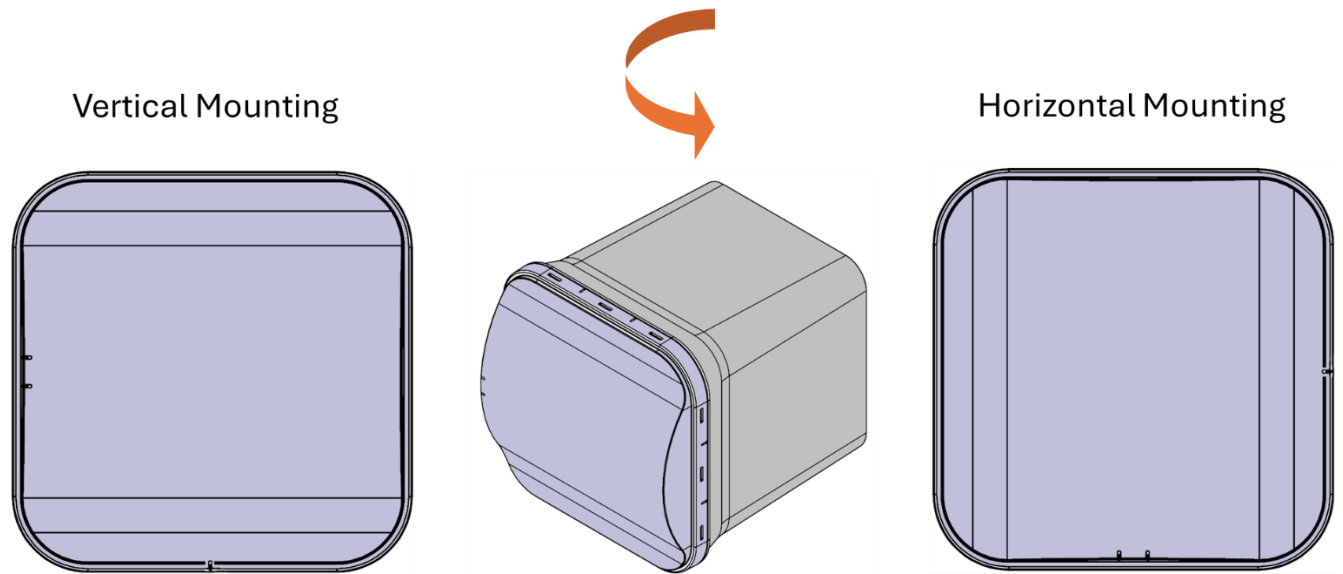
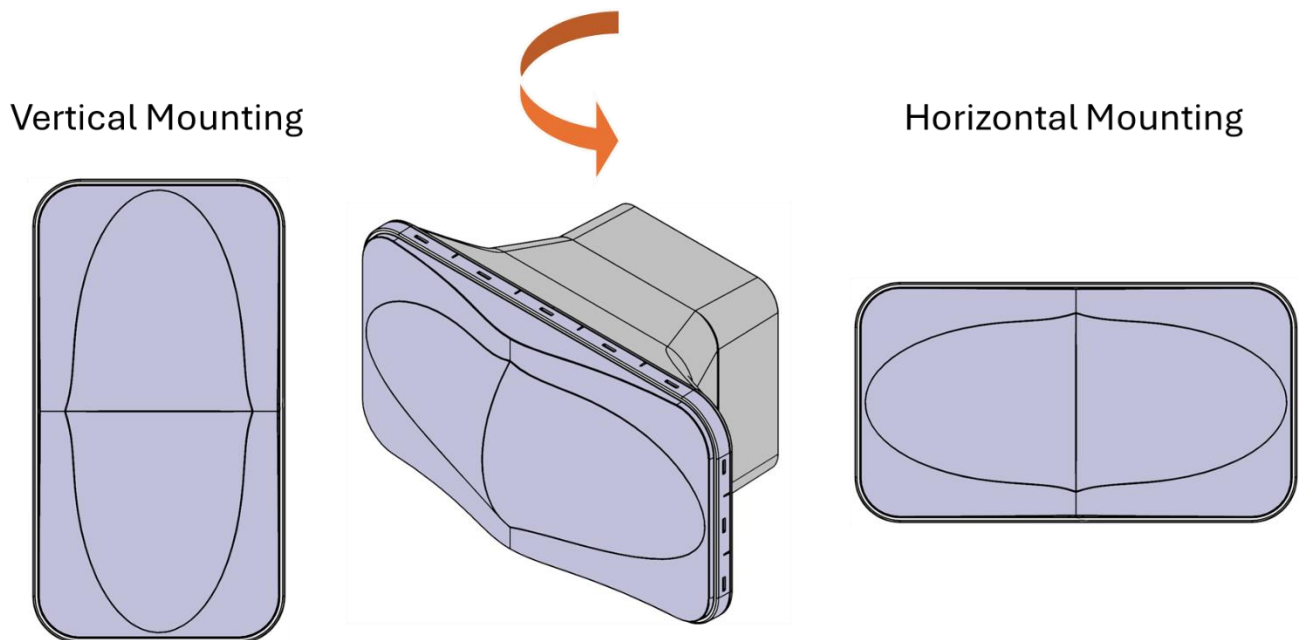
RADWIN TerraNet lens antennas could be used horizontally or vertically like described below. This positioning does not need any specific software configuration; the system automatically adapts to the position of the antenna.



Both models of antenna come with pre-drilled holes to match both positioning vertical and horizontal. Check that the holes are aligned between the antenna and the TerraNet V90 device enclosure before screwing them together.



When attaching the bracket to the enclosure using the 4 x M4x10 bolts, please do not overtighten. We recommend to use a torque of 1Nm.

45° antenna lens positioning**90° antenna lens positioning****3.3.6 Connect the device**

First, Connect CAT-5e / DC cables to the POE.

Using V40 and V120 Sealing Kit

TerraNet V40 and V120 could get an improved IP (IP54 to IP56) using an external sealing kit to protect the connectors area.



The Sealing Kit cannot be used if you plan to use both connectors ETH0 and ETH1 (for example for failover or PoE out).

The sealing kit is made of silicone. It provides better closure to the connectors area to ensure water will not pass through.



Here are the different steps to follow-up in order to correctly install the sealing kit into the device:

1. Pass the ethernet cable through the X cut into the sealing kit.



The Sealing Kit cannot be used with cables having a diameter larger than 6.9mm.

2. Move the cable to the hole and pass a part of the cable through it.



3. Connect the cable to the device on ETH) PoE in port.



4. Move the sealing kit inside the connector area and push it to the end. To avoid damaging the cable, you need to maintain the cable in the right position during the process.



5. Insert the cover to close the connectors area.



6. Close the cover. The device is ready for power up.



Connecting the cable on V90

Open the gland, insert the cable in the Ethernet port and close the gland to protect connectors from water and dust.

Using additional ports on V40 and V120

If you plan to power the device using 48V DC or, to use failover or, if you want to power an additional device using the PoE out of the TerraNet V40 or V120, you need first to remove the flaps.



Figure 15: Remove the flaps

Power Up the device

- To power up the unit, connect the PoE adaptor to the power.



Figure 16: Power up the device through the PoE

- Check the power LED on the PoE, if the LED is green, go and check the status LEDs on the device.

3.4 Powering up the device

3.4.1 Input Power



Note

- The input power should be +48VDC with a minimum of .5A (24W).
- TerraNet device support active PoE (802.3at, etc).
- -48VDC is **not** supported.

TerraNet devices can be powered by two ways:

1. The Power over Ethernet (PoE) port or via
2. The DC input terminal (**ONLY for V40 and V120**).



Warning

Both powering methods should not be used at the same time.

The input power should be +48VDC in both cases with a minimum of .5A (24W).

Although it is not recommended, a lower input voltage may be used. However, there are two main limitations when using a lower input voltage:

- The RJ45 cable should be as short as possible and the use of a cable such as CAT6+ is highly recommended. On longer RJ45 cable, the voltage drop will be higher due to the higher current needed to meet the needs of the radio.
- The PoE out current is limited to .5A. At 38VDC, there should be sufficient power for a second radio unit depending on the length of your RJ45 cable.

3.4.2 Output Power

TerraNet radio units can send passive PoE out of the 1Gbps port to power another device using the input power provided to the device. These settings can be found under the Network/Ethernet section of the Web UI – Please refer to the User Manual.

The output voltage is the same as the input voltage and delivers a maximum current of .5A.



When using the PoE out feature on the TerraNet radio unit, you should take care there is adequate power supply to power out a secondary device. The supplied PoE injector may not be adequate to power both devices and may result in unstable operation.



When connecting a second device (TerraNet or any other POE in device) to the TerraNet radio unit power out, **we recommend using our RW-9921-1059 POE.**



Passive PoE out on the 1Gbps port can damage non-PoE devices. Do not connect non-PoE devices to the 1Gbps port when PoE out is enabled.

3.5 LED Behavior



Please be patient the first time you turn on your device, as it may take more time than normal to boot up while the device flashes its internal modem firmware. Three blinking signal LEDs will indicate that this process is currently in progress and should last about 30 seconds to 1 minute.

3.5.1 V40 and V120 Signal Level LEDs

Signal LEDs are updated every four seconds.

Behavior	Meaning
All signal LEDs are off	The device is currently operating in Access Point Mode, or the device is powered off.
The first LED is on & slowly blinking	The device is in Station Mode and currently scanning, or device is in Access Point Mode waiting for the Station to connect.
All three LEDs are on & fast blinking	The device is currently updating its modem firmware. This only happens at boot after a firmware upgrade. Warning: A normal upgrade should take less than 1 minute, but if the modem firmware is being recovered from an invalid state, this process could take up to 10 minutes.
The first LED is on and not blinking	The device is in Station Mode and currently connected to a peer with low RSS.
The first two LEDs are on, not blinking	The device is in Station Mode and currently connected to a peer with medium RSS.
All three LEDs are on, not blinking	The device is in Station Mode and currently connected to a peer with high RSS.

3.5.2 V90 Signal Level LED

Signal LED is updated every four seconds.

Behavior	Meaning
Signal LED is off	The device is currently operating in Access Point Mode, or the device is powered off, or the modem is not detected.
Signal LED is blinking slowly	The device is in Station Mode or in PtP mode and scanning.
Signal LED is blinking at a medium speed	The device is in Station Mode or in PtP mode and connected to a peer with low RSS.
Signal LED is blinking at a fast speed	1st Use Case: If the unit is just booting up, the device is flashing the internal modem firmware. 2nd Use Case: If the unit is fully booted, then the device is in Station mode or in PtP mode and connected to a peer with a medium signal strength.
Signal LED are on, not blinking	The device is in Station Mode or in PtP mode and connected to a peer with high RSS.

3.5.3 Ethernet Port LEDs

2.5G ETH0: The link light will turn on when a link is established, and the power light will turn on once the device is supplied with power.

1G ETH1: The link light will turn on and blink when a link is established, and the PoE out light will turn on when PoE out is enabled in software.

3.6 Initial Configuration

3.6.1 Connect the device

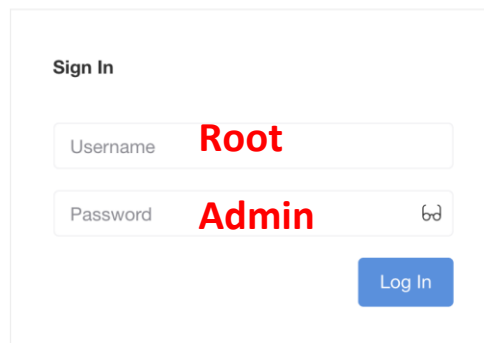
Power on the device. You can do this by using the DC input terminal or by using the included PoE adapter. If using the PoE adapter, plug the power cable attached to the PoE adapter into the wall. Next, plug one side of an ethernet cable into the adapter's POE port, and the other into the device's 2.5G ETH0 PoE-in port.



We recommend to use an outdoor lightning protection when connecting the device through its ETH ports – **RADWIN LPU Reference: RW-9924-0106/7** .

Connect your laptop's ethernet cable into either the data port of the PoE adapter (2.5G) or the ETH1 (1G) port on the device, to give your laptop or device connectivity. By default, DHCP client is enabled on the main management network bridge of the device. If your device cannot obtain an IP from an upstream DHCP server, it will fall back to 10.0.0.120, unless otherwise configured.

Log in to your device's local web UI in your web browser at the DHCP-assigned IP, or the fallback IP previously mentioned. Login to the device:



The screenshot shows a web browser window with the RADWIN login page. The page has a white background with a light gray border. At the top, it says "Sign In". Below this, there are two input fields. The first is labeled "Username" and contains the text "Root" in red. The second is labeled "Password" and contains the text "Admin" in red. To the right of the password field is a small eye icon. Below the input fields is a blue button labeled "Log In".

Figure 17: Login – Default Username and password



The device's default fallback IP is 10.0.0.120, we recommend you change the device's default user credentials after you log in for the first time.

Go to the configuration settings tab via the main nav menu on the left to configure your device. Please refer to the User Manual for details about each setting.

3.6.2 Link Establishment



You will automatically be logged out of your session if you're inactive for more than 30 minutes.

To establish a link between ODUs, you need to activate the ODU by configuring essential parameters such as link ID, password, and antenna parameters for external ODUs, operating band and channels in the Hub).

By default, TerraNet V120 and V90 radio units are configured to operate in Access Point Mode, and TerraNet V40 radio units are configured to operate in Station Mode with the same SSID. Both are configured to use by default 'passphrase' as the encryption key.

This means once you power up the devices within range of each other, they should link up without any additional configuration (default channel is 5), and immediately start passing traffic.



Remember that 60GHz frequency requires LOS and is affected by rain fade.

For minimal setup, it is recommended to change the SSID to a unique identifier and to set a unique encryption key. To do so, In the Web UI, Wireless → 60 GHz Radio Settings, and input the new settings on both devices (and subsequent client radios). Click save to apply the changes. If the channel width or operation mode has been changed, it will require the device to reboot to take effect.



For the devices to link together, the SSID, channel width, and encryption type/key must all match.

When you aim one device at the second one, they should connect and pass traffic immediately. In close range such as lab setups, it may be difficult to reach full performance due to the strength of the signal and high-power reflections caused by walls and other objects.

You can see the status of the links between the AP and client on the dashboard page of the Web UI.

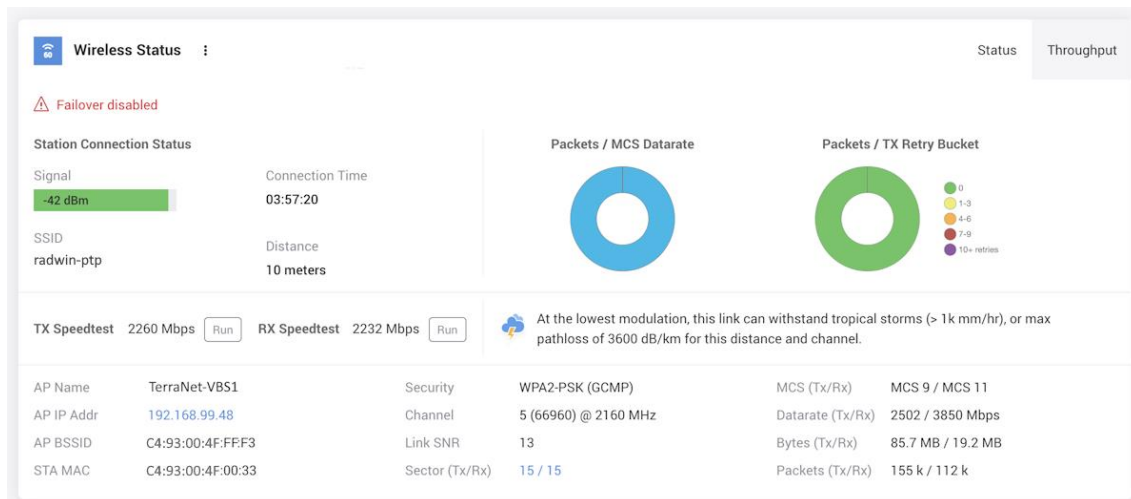


Figure 18: See device's wireless status

3.6.3 V90 Alignment Recommendations

When using V90 with an external antenna, the elevation width of the device is reduced. To reach the maximum performance with the device, you should follow the following steps for PtP or PtMP deployment.

PtP (perform at both sides of the link simultaneously)

1. Connect the radio to power.
2. Set the mounting kit towards the remote site (azimuth, elevation). You could use a laser or any other tool to help you, improving the alignment.
3. Login to WebUI and under "configuration" tab select the "Wireless page" and change the "Lock to Boresight Beam" status to ON.

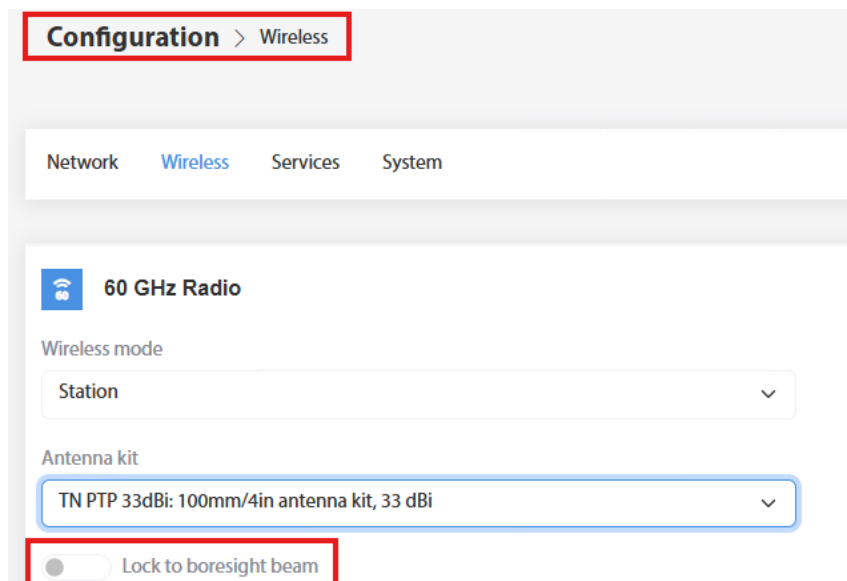


Figure 19: PtP: Lock the boresight beam

4. Go to the “Tools” tab and select the “Site Survey & Aiming” page.

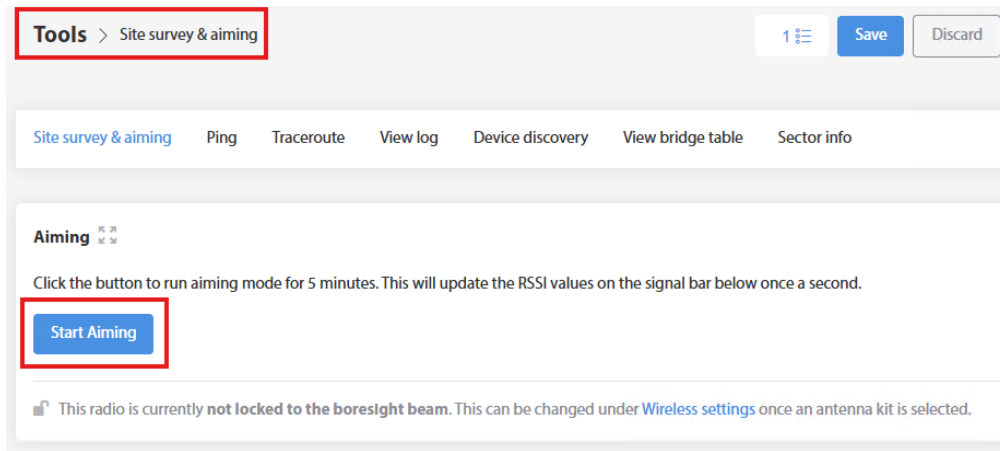


Figure 20: PtP: Aiming to the second device

- a. Click on the Start Aiming button at “Client” side of the link and wait for 2 minutes before changing the device alignment.
- b. Start rotating the antenna alignment on the Azimuth to the Right/Left in small steps – wait 3 seconds before every change.
- c. Someone needs to read RSSI levels for the installer. RSSI levels are updated every 3 seconds.
- d. When you acquire a signal, you could compare it to the expected LBC results and continue aligning the Azimuth until the best signal is achieved.
- e. Lock the mounting kit Azimuth movement.
- f. Start rotating the “Client” antenna alignment on the Vertical Uptilt/Down-tilt in small steps – wait 3 seconds before every change.
- g. Someone needs to read RSSI levels for the installer.
- h. Lock the Mounting kit Elevation movement when the desired RSSI achieved.
- i. When required, repeat the Aiming process at the “AP” side of the link – start from the Azimuth alignment and then on the Elevation.

When desired RSSI achieved at both sides on the link, disable the “Lock Boresight Beam” to enable the Beamforming mechanism to start.

On the dashboard, the target RSSI provides the best possible RSSI the device could obtain when alignment is done perfectly.

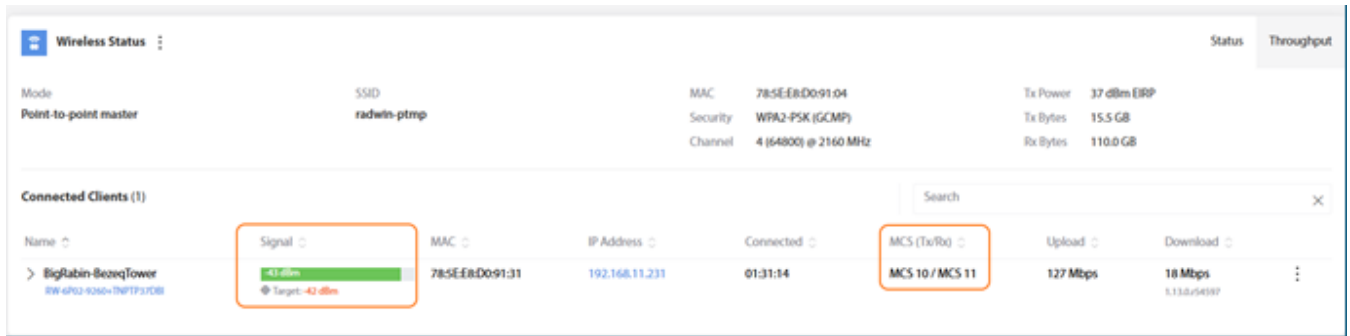


Figure 21: PtP: Target aligned with real RSSI

PtMP

AP side

1. Connect the “AP” radio to power.
2. Set the mounting kit to the planned azimuth.
3. Set the mounting kit to the planned Down-tilt/Up-tilt.

Client side

1. Connect the radio to power.
2. Set the mounting kit towards the AP site.
3. Login to WebUI and under “configuration” tab select the “Wireless page” and change the “Lock to Boresight Beam” status to ON.

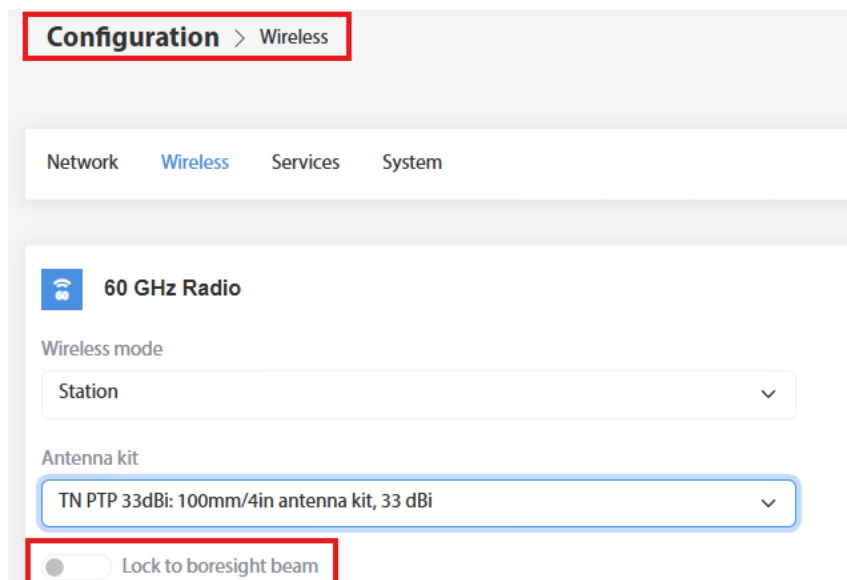


Figure 22: PtMP: Lock the boresight beam

4. Go to the “Tools” tab and select the “Site Survey & Aiming” page.

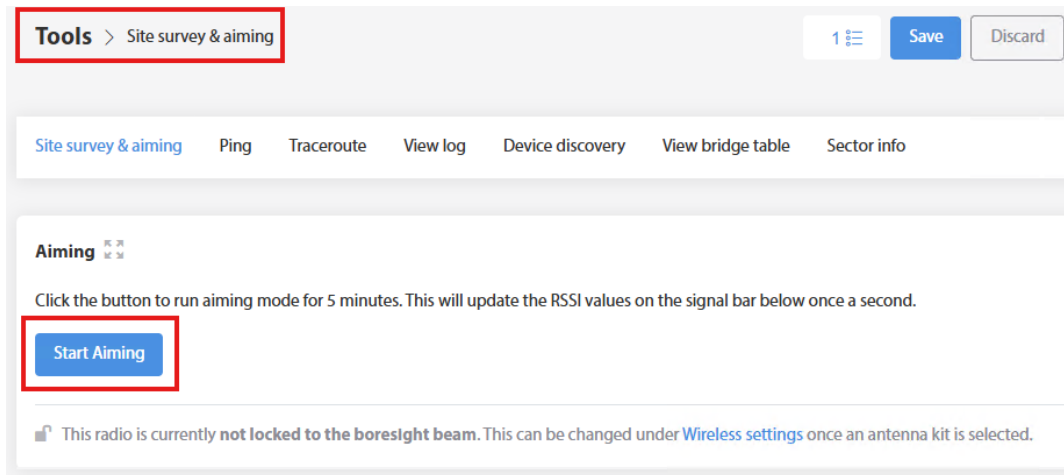


Figure 23: PtP: Aiming the device

- i. Click on the Start Aiming button at “Client” side of the link and wait for 2 minutes before changing the device alignment.
- ii. Start rotating the antenna alignment on the Azimuth to the Right/Left in small steps – wait 3 seconds before every change.
- iii. Someone needs to read RSSI levels to the installer.
- iv. When you see a signal, compare it to the expected LBC results and continue aligning the Azimuth until the best signal is achieved.
- v. Lock the mounting kit Azimuth movement.
- vi. Start rotating the “Client” antenna alignment on the Vertical Uptilt/Down-tilt in small steps – wait 3 seconds before every change.
- vii. Someone needs to read the RSSI levels to the installer.
- viii. Lock the Mounting kit Elevation movement when the desired RSSI achieved.

4 Regulations

4.1 FCC/IC Warning

Any Changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

This device complies with part 15 of the FCC Rules. Operation is subject to the following two conditions:

1. This device may not cause harmful interference, and
2. This device must accept any interference received, including interference that may cause undesired operation.



This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to part 15 of the FCC Rules.

These limits are designed to provide reasonable protection against harmful interference in a residential installation.

This equipment generates uses and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation.

4.2 ISEDC Warning



This radio is intended for use in occupational/controlled conditions where users have full knowledge of their exposure and can exercise control over their exposure to meet the occupational limits in FCC/ISEDC and International standards. This radio device is not authorized for general population consumer use.

4.2.1 RF Exposure Requirements

To control your exposure and ensure compliance with the occupational/controlled environmental exposure limits, always adhere to the following procedures.

4.2.2 Guidelines

User awareness instructions should accompany the device when transferred to other users. Do not use this device if the operational requirements described herein are not met.

4.2.3 Operating Instructions

Keep the radio unit at least 2.5 cm away from the face. Keeping the radio at the proper distance is important as RF exposure decreases with distance from the antenna. The antenna should be kept away from the face and eyes.

Use of non-approved antennas, batteries, and accessories causes the radio to exceed the FCC/ISED RF exposure guidelines.

Any changes or modifications not expressly approved by the party responsible for compliance could break the user's authority to operate the equipment.

This device complies with Innovation, Science and Economic Development Canada license-exempt RSS standard (s). Operation is subject to the following two conditions:

1. this device may not cause interference, and
2. This device must accept any interference, including interference that may cause undesired operation of the device.

Tout changement ou modification non expressément approuvé par la partie responsable de la conformité pourrait annuler l'autorité de l'utilisateur à faire fonctionner l'équipement.

Cet appareil est conforme aux normes RSS exemptes de licence d'Innovation, Sciences et Développement économique Canada. Le fonctionnement est soumis aux deux conditions suivantes : (1) cet appareil ne doit pas causer d'interférences, et (2) cet appareil doit accepter toute interférence, y compris les interférences pouvant entraîner un fonctionnement indésirable de l'appareil.

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